McMurdo Station, Ross Island

BUILDING INFORMATION SHEET

Dorm

Location ID: B209

Three-story personnel dormitory housing up to 134 persons. The first floor includes 17 two-person rooms, a lounge, a public restroom, and a sauna. The second floor includes 24 two-person rooms and a small lounge. The third floor includes 26 two-person rooms.

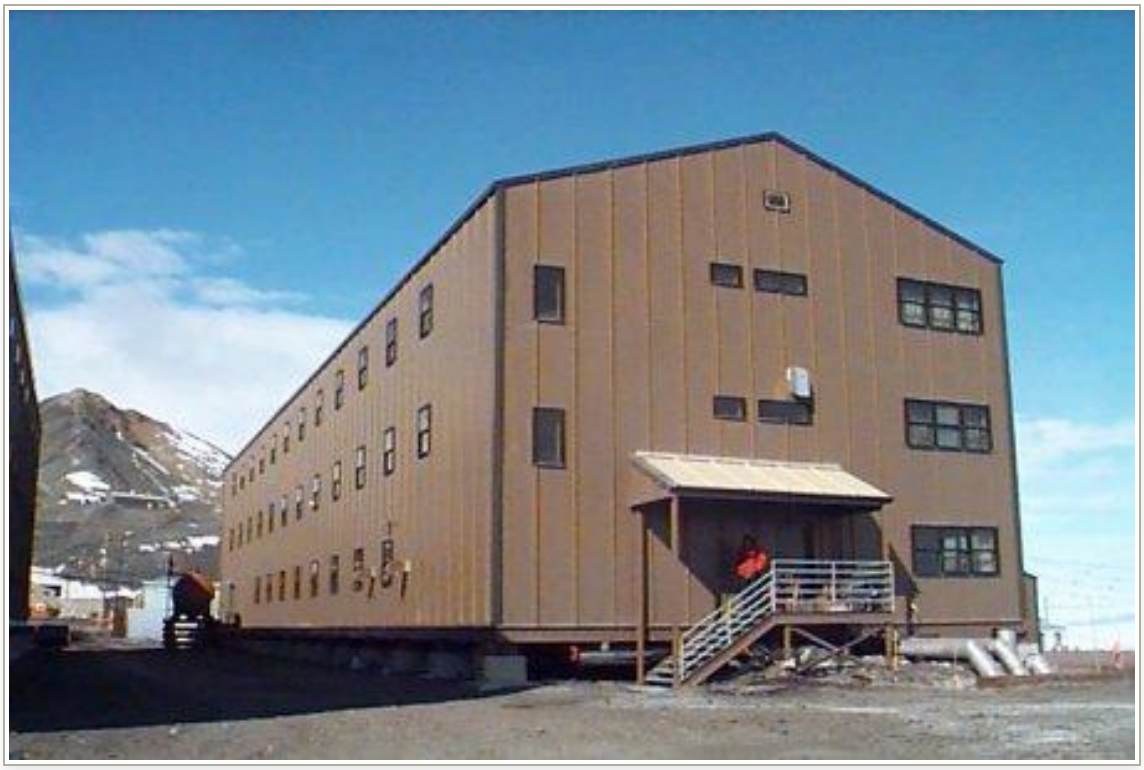


Photo 1: *Dorm*

General Structure Information

Usage	Dormitory
Occupancy Group	R-1 Dormitories
Maximum Occupancy	381
Construction Type	Type II, 1-Hour, non-combustible; steel-framed structure with 2-1/2" thick foam-insulated metal siding and roofing, steel-framed heavy timber first floor, steel-framed metal second and third floors. Steel-stud gypsum board interior partitions.
Year Constructed	1988-89
Dimensions	48' X 168'

Floor Area

Floor	Area
Floor 1	8,064 sq ft
Floor 2	8,064 sq ft
Floor 3	8,064 sq ft
Total	24,192 sq ft

HVAC System

Heating	In 1999 the York Shipley oil-fired glycol boiler was replaced with three 330,000 Btu/hr input, oil-fired, cast iron, Hydrotherm glycol boilers. The boilers are controlled in a staged manner so that only the number of boilers required will activate. The heating piping in the mechanical room was also repiped in 1999. The system is configured to a primary-secondary heating system with the distribution piping in a reverse return configuration. The primary loop includes the boiler and a loop to the heat exchanger for the potable hot water. The temperature setpoint for the primary loop is 180F. The secondary loop provides heat to the baseboard radiators on six different zones and to the two air handling units on one zone. The temperature setpoint for the secondary zone varies with the outdoor temperature. The range is approximately 100F to 180F. Heat is supplied from the primary loop to the secondary loop by means of a diverting valve.
Air Handling Units	Two Trane, Climate Changer air handling units. They are located at each end of the building in the attic vestibule. They supply fresh air only at 60 degrees F.
Fuel Tank	2,500 gallon tank

Water and Sewer

Water Service	2 inch
Sewer Service	4 inch
Connections	Sinks: Toilets: Urinals: 1 Showers: Water Fountains: 1 Washing Machines: 6 Ice Machine: 2

	Sauna: 1 100 lines - pending verification for all buildings
Water Heater	Glycol heat exchanger with 440 gallon hot water storage tank

Electrical Service

Service Feeder	Feeder A
Electrical Service	208/120V, 3PH, 400A main
Phone and LAN	100 lines - pending verification for all buildings

Fire System

Fire Panel Type	Pyrotronics System 3 control panel, King-Fisher KFRTI-WP-8 radio interface/transmitter panel, pull stations, alarm horns
Suppression Type	Pre-action sprinkler system connected to Firetrol pump and resevoir in Building 151 and equipped with Star, 6-inch, model D, electromagnetic deluge valve with air pressure supervision. Deluge valve releases upon low air pressure and heat detector actuation. Firetrol pump starts on water pressure loss in main trunk line to dorms.

Floorplans

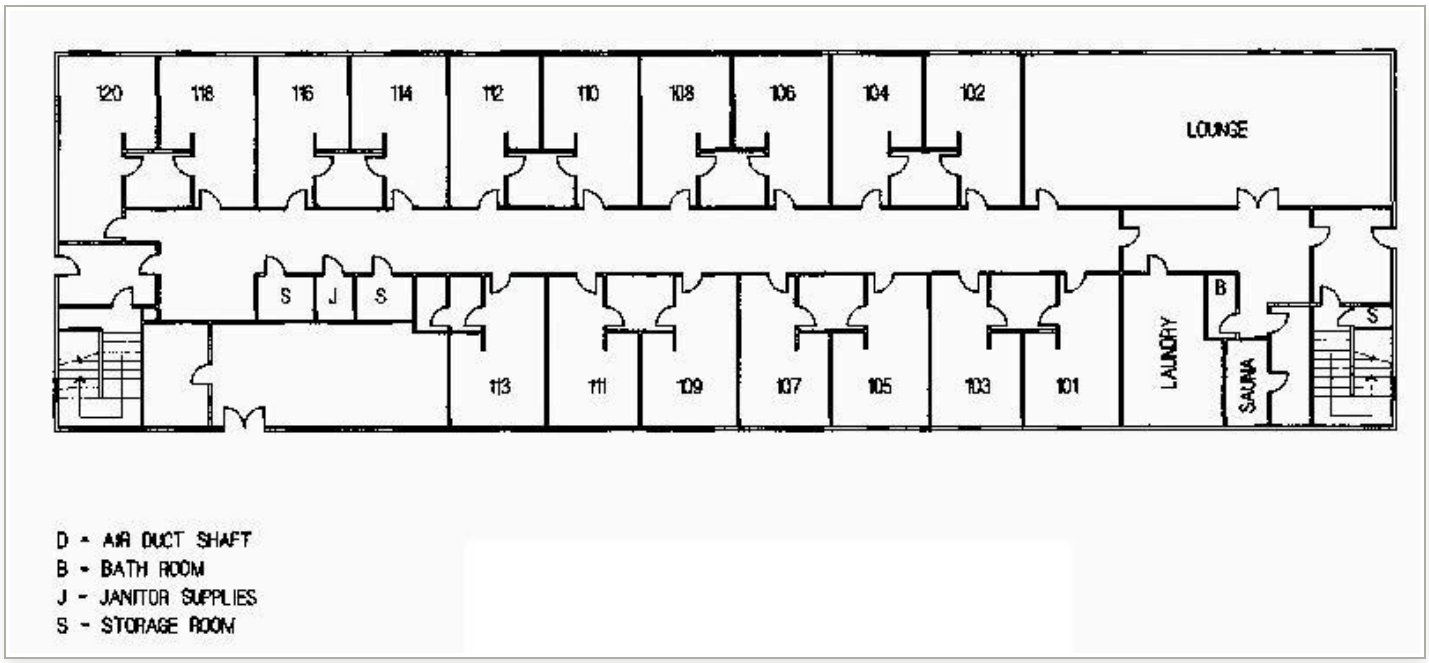


Figure 1: *Dorm*

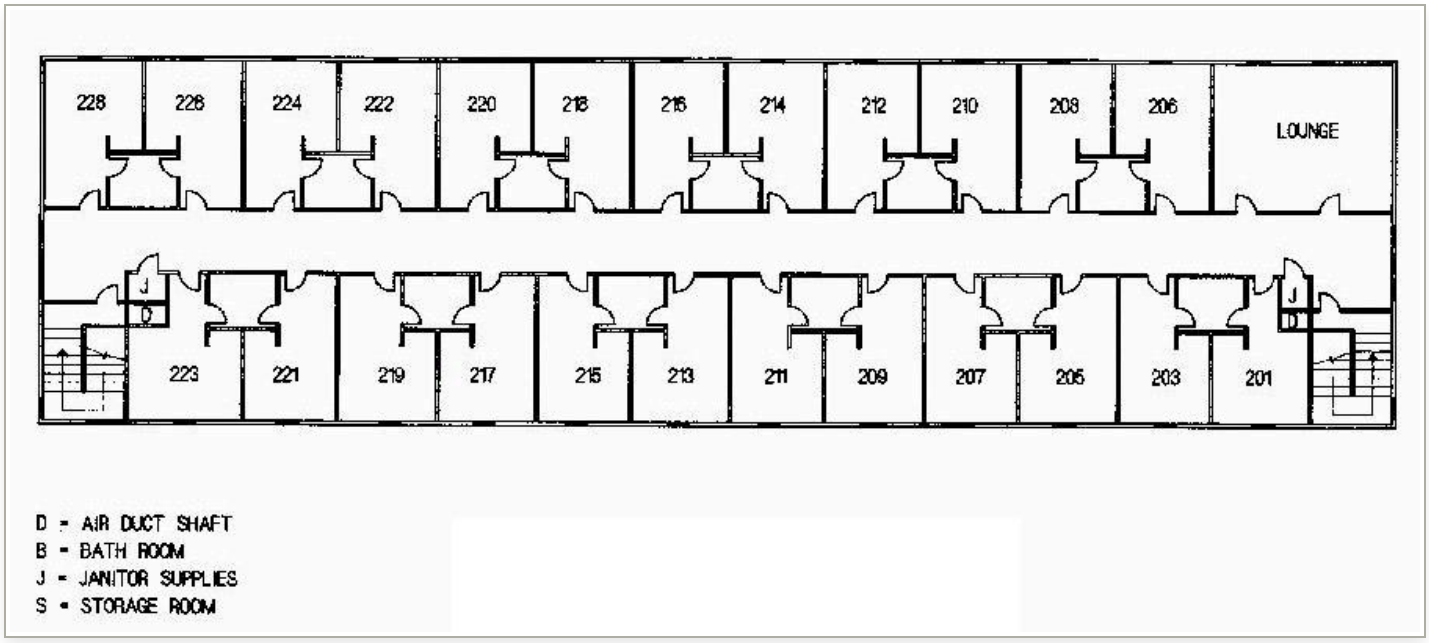


Figure 2: *Dorm*

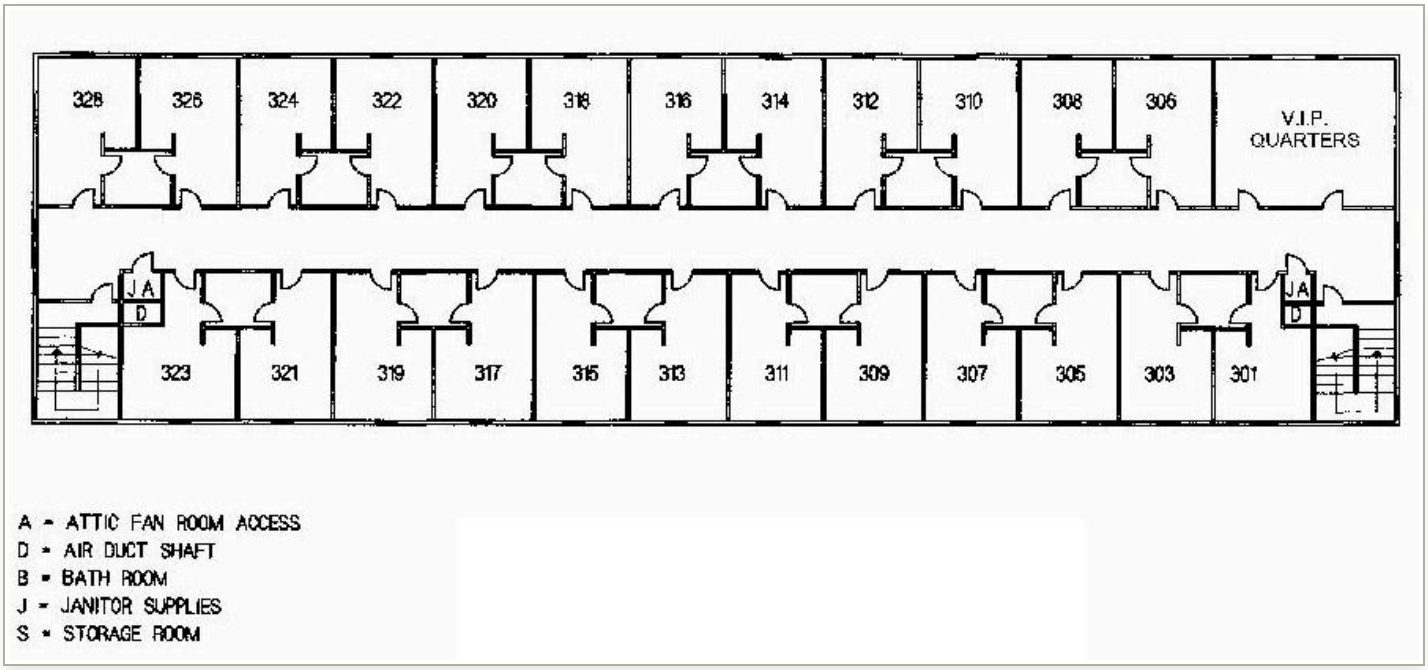


Figure 3: *Dorm*